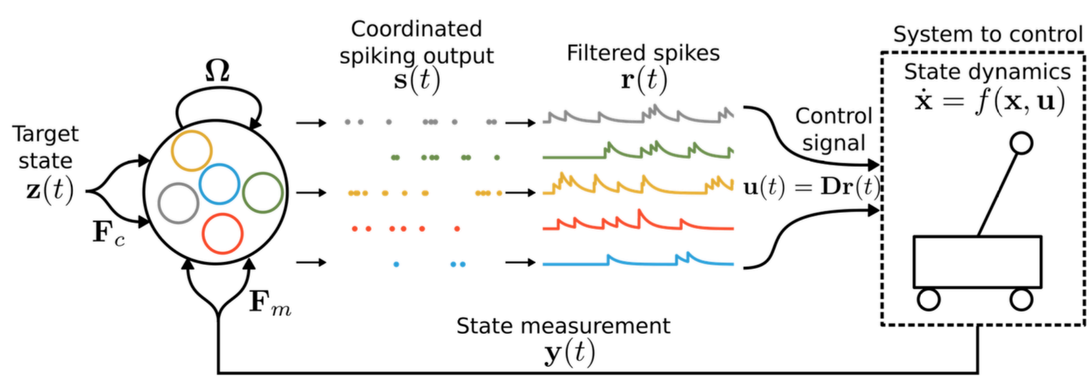
Motivation

- Leaky Integrate-and-Fire Neurons
  - work by integrating (accumulating) input currents over time to increase a "membrane potential" and "fire" (produce a spike) when this potential crosses a threshold, similar to how a leaky RC circuit charges. After firing, the potential resets. The "leaky" part refers to the fact that the membrane potential also gradually decays or "leaks" back to a resting value over time unless it is being driven by input
- The SCN idea
  - a population of LIF neurons only fire when it reduces the global representation error. This yields spare, irregular, yet coordinated spikes
  - The paper shows that the same network can implement a linear dynamical system (add a “slow” recurrent term).
  - Kalman filter for optimal state estimation through noisy observations
  - LQR for optimal control to a target z(t)
  - The control u is a linear readout of the spikes
  - -> Spiking LQG

Setup



- **Fig. 1:** Controlling dynamical systems with spiking neural networks. A recurrent spiking neural network (left) will emit spikes based on its inputs and connectivity, which are translated into a control signal to control some target system (right). $\mathbf{F}_c$ and $\mathbf{F}_m$ represent input weights, $\mathbf{\Omega}$ represent recurrent weights, and $\mathbf{D}$ represent read-out weights. By first defining the optimal control solution for the system, and then directly translating the resulting control parameters into these connection weights, we can generate a robust and population-wide coordinated spiking network that accurately controls identified systems.
- x: state
- y: incomplete measurement of the system state
- z: incomplete measurement of the target state
- u: control signal
- s: spiking signal emitted from a recurrent SNN, used to control the stater: filtered spikes

Emitted Spikes and Filtered Spikes

- s: spiking signal emitted from a recurrent SNN, used to control the state
  - $$s_i(t) \;=\; \sum_k \delta\big(t - t_i^{(k)}\big), \qquad s(t) = \big[s_1(t), \dots, s_N(t)\big]^\top.$$
  - discrete: $$s_i[n] = \begin{cases} \frac{1}{\Delta t}, & \text{if neuron } i \text{ spikes in bin } n, \\ 0, & \text{otherwise,} \end{cases}$$
  - s_i(t) is the sum of the spikes emitted from the neurons in time t.
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    - Each $t_i^{(k)}$ is the k-th spike time of neuron i.
    - Integrating s_i over any interval counts spikes: $\int_{t_0}^{t_1} s_i(t)\, dt = \#\{\text{spikes of } i \text{ in } [t_0, t_1]\}$.
    - This is a **distribution** (generalized function), not an ordinary continuous signal. In their LIF network, a spike occurs when the voltage $v_i(t)$ hits its threshold T_i; then s_i contributes a delta at that instant.
- r: filtered spikes
  - $$\dot r(t) = -\lambda r(t) + s(t) \quad \Rightarrow \quad r = \big(e^{-\lambda t} H(t)\big) * s,$$ so each spike adds a unit jump to r that then decays with time constant $1/\lambda$. Closed-form control with spike ...
  - $$\dot r(t) \;=\; -\lambda\, r(t) + s(t) \quad \Longleftrightarrow \quad r(t) = \int_{-\infty}^{t} e^{-\lambda(t-\tau)}\, s(\tau)\, d\tau.$$
- Dirac impulse delta function
  - $$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \quad \text{with} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1.$$
  - it's not a real function, so approximated with limits: $\delta_a(x) = \frac{1}{|a|\sqrt{\pi}} e^{-(x/a)^2}$ a->0

Spike Coding Theory

- If a K-dimensional state x(t) ∈ R^K is fully observable without noise, SCN theory defines how a recurrent SNN can optimally track this signal
  - Assumptions
    - the input signal estimate $\hat{x}$ can be linearly decoded from the spiking activity as $\hat{x} = Dr$, where D in R^(N x K) are known decoding weights (in this paper randomly drawn from the normal distribution and normalized)
    - spikes should only be emitted when this improves the coding error (GREEDY SPIKING RULE): $||x - Dr||_2^2 > |x - Dr - D_i|_2^2$

- The main equations
    - $\hat{x} = Dr$
    - $\dot{r} = -\lambda r + s$
    - $\Omega_f = -D^T D$
    - $\dot{v} = -\lambda v + D^T(\dot{x} + \lambda x) + \Omega_f s$